Energy-consistent anelastic and compressible box models: continuous equations and numerical scheme

Notes on the `TraceableEnergetics` code (AN2D and FC2D)

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Conventions. AN2D denotes the 2D anelastic model and FC2D the 2D fully compressible box model. Part I contains only continuous physics; Part II specifies the discrete scheme completely enough to be re-implemented, and derives its conservation properties. Where this document and any other write-up disagree, it follows the code. The correspondence between mathematical symbols and code names is given in Table 1.

Part I
Continuous physics

1 Setting and variables

Both models are posed in a vertical box $(x, z) \in [0, L_x] \times [0, L_z]$ with z upward and no dependence on y. The velocity is $\mathbf{u} = (u, w)$ and $\boldsymbol{\omega} = \omega\, \hat{\mathbf{y}}$ with

$$\omega = \partial_z u - \partial_x w. \tag{1}$$

Gravity acts along $-\hat{\mathbf{z}}$ with constant magnitude $g > 0$. The geopotential is

$$\Phi(z) = g\,(z - z_0), \qquad \boldsymbol{\nabla}\Phi = g\,\hat{\mathbf{z}}, \tag{2}$$

with z the height above the bottom of the box and z_0 the height at which Φ vanishes (`space.z0`, by default the bottom, $z_0 = 0$). The origin of Φ is arbitrary: shifting it changes the total energy by a constant times the (conserved) mass.

The fluid is a binary mixture (solvent and solute) with mass fraction q of solute and an entropic variable ϑ that is a function of the specific entropy s and of q only, $\vartheta = \vartheta(s, q)$, and is therefore conserved by adiabatic, isohaline motion. In the experiments ϑ is the potential temperature; any entropic variable is allowed (e.g. $\vartheta = s$).

2 Thermodynamics

2.1 Fundamental relation

The fluid is specified by its specific enthalpy $h(P, \vartheta, q)$, with

$$\mathrm{d}h = v\,\mathrm{d}P + h_\vartheta\,\mathrm{d}\vartheta + h_q\,\mathrm{d}q, \qquad h_\vartheta = \left.\frac{\partial h}{\partial \vartheta}\right|_{P,q}, \quad h_q = \left.\frac{\partial h}{\partial q}\right|_{P,\vartheta}, \quad v = \left.\frac{\partial h}{\partial P}\right|_{\vartheta,q}. \tag{3}$$

The internal energy is $e = h - Pv$. The Bernoulli potential is

$$B = h - \vartheta\, h_\vartheta - q\, h_q. \tag{4}$$

From (3), $\mathrm{d}B = v\,\mathrm{d}P - \vartheta\,\mathrm{d}h_\vartheta - q\,\mathrm{d}h_q$, hence the pressure force can be written without $\boldsymbol{\nabla}P$:

$$-v\,\boldsymbol{\nabla}P = -\vartheta\,\boldsymbol{\nabla}h_\vartheta - q\,\boldsymbol{\nabla}h_q - \boldsymbol{\nabla}B. \tag{5}$$

Also, for $\rho e = \rho h - P$ regarded as a function of $(\rho, \rho\vartheta, \rho q)$,

$$\mathrm{d}(\rho e) = B\,\mathrm{d}\rho + h_\vartheta\,\mathrm{d}(\rho\vartheta) + h_q\,\mathrm{d}(\rho q). \tag{6}$$

Indeed $\mathrm{d}(\rho h - P) = h\,\mathrm{d}\rho + \rho\,(v\,\mathrm{d}P + h_\vartheta\,\mathrm{d}\vartheta + h_q\,\mathrm{d}q) - \mathrm{d}P = h\,\mathrm{d}\rho + \rho h_\vartheta\,\mathrm{d}\vartheta + \rho h_q\,\mathrm{d}q$ (using $\rho v = 1$), and $\rho\,\mathrm{d}\vartheta = \mathrm{d}(\rho\vartheta) - \vartheta\,\mathrm{d}\rho$, likewise for q. This is what makes $(\rho, \rho\vartheta, \rho q)$ the natural prognostic variables for energy conservation.

2.2 Temperature, entropy and chemical potential

The temperature and entropy are defined by $\mathrm{d}h = T\,\mathrm{d}s + v\,\mathrm{d}P + \mu\,\mathrm{d}q$, i.e. $T = \partial h/\partial s|_{P,q}$ and $\mu = \partial h/\partial q|_{P,s}$. Since $s = s(\vartheta, q)$ does not depend on P, comparing with (3) at fixed P gives

$$T\,\mathrm{d}s + \mu\,\mathrm{d}q = h_\vartheta\,\mathrm{d}\vartheta + h_q\,\mathrm{d}q \quad \Longrightarrow \quad s_\vartheta := \left.\frac{\partial s}{\partial \vartheta}\right|_q = \frac{h_\vartheta}{T}, \qquad s_q := \left.\frac{\partial s}{\partial q}\right|_\vartheta = \frac{h_q - \mu}{T}, \tag{7}$$

Symbol	Meaning [unit]	Code
L_x, L_z, g	box size [m], gravity [$\mathrm{m\,s^{-2}}$]	`space.Lx`, `space.Lz`, `space.g`
z_0	height above the bottom where $\Phi = 0$ [m]	`space.z0`
ρ, $v = 1/\rho$	density [$\mathrm{kg\,m^{-3}}$], specific volume	`m[:,:,1]` (FC2D)
P	pressure [Pa]	`p`
p	reduced pressure (AN2D), $p = (P - P_R)/\rho_R$ [$\mathrm{J\,kg^{-1}}$]	`scratch.phi`
ϑ, q	entropic variable, solute fraction [$\mathrm{kg\,kg^{-1}}$]	`consvar`, `q`
$h(P, \vartheta, q)$	specific enthalpy [$\mathrm{J\,kg^{-1}}$]	`specific_enthalpy`
$h_\vartheta = \partial h/\partial\vartheta\|_{P,q}$	conjugate variable of ϑ	`conjvar`
$h_q = \partial h/\partial q\|_{P,\vartheta}$	modified chemical potential [$\mathrm{J\,kg^{-1}}$]	`chempot`
$\mu = \partial h/\partial q\|_{P,s}$	chemical potential [$\mathrm{J\,kg^{-1}}$]	(enters only through Γ)
T	temperature [K]	`T`
c_p	heat capacity [$\mathrm{J\,kg^{-1}K^{-1}}$]	`Cp`
$B = h - \vartheta h_\vartheta - q h_q$	Bernoulli potential [$\mathrm{J\,kg^{-1}}$]	`B`
b	buoyancy (AN2D) [$\mathrm{m\,s^{-2}}$]	`buoyancy`
$K = \lvert\mathbf{u}\rvert^2/2$	kinetic energy per unit mass	`K`
$\boldsymbol{\tau}$, η	viscous stress, dynamic viscosity [Pa s]	`dyn_visc`
κ_T, κ_q	diffusivities [$\mathrm{m^2\,s^{-1}}$]	`k_T`, `k_q`
Γ	barodiffusion coefficient [$\mathrm{Pa^{-1}}$]	`Gamma`
$\partial\mu/\partial P\|_{T,q}$, $\partial\mu/\partial q\|_{P,T}$	derivatives defining Γ	`chemical_potential_derivatives`
$\partial\vartheta/\partial q\|_{P,T}$	used in the flux of ϑ	`dcons_dq`
$\mathbf{J}_\vartheta, \mathbf{J}_q$	diffusive fluxes of ϑ, q	`consflux`, `qflux`
σ_ϑ	production of $\rho\vartheta$	`sigma_cons`
$\rho\varepsilon$	viscous dissipation per unit volume	`rho_eps`
$\rho_R(z), P_R(z)$	anelastic reference density, pressure	`rho_R`, `p_R`

Table 1: Mathematical symbols and code names (Greek letters in the source are spelled out). The capital P is the full pressure; p is the reduced pressure of the anelastic model. The fluid (equation of state) is supplied by the `ClimFluids` library; AN2D is the package `CFAnelastics` and FC2D the package `CFCompressible`. The flux laws of §3 are the functions `salt_flux`, `heat_to_consvar` and `consvar_flux`, and the schemes κ_T, κ_q, η the types `ThermoFluxConsistent`, `ThermoFluxDowngradient` and `ViscosityScheme`, all of `CFDiffusionSchemes`.

both functions of (ϑ, q) only. By the chain rule on $h(P, \vartheta(P, T, q), q)$,

$$\left.\frac{\partial h}{\partial q}\right|_{P,T} = h_\vartheta \left.\frac{\partial \vartheta}{\partial q}\right|_{P,T} + h_q. \tag{8}$$

The Gibbs function $G = h - Ts$ satisfies $\mathrm{d}G = -s\,\mathrm{d}T + v\,\mathrm{d}P + \mu\,\mathrm{d}q$, hence the Maxwell relation $\partial\mu/\partial T|_{P,q} = -\partial s/\partial q|_{P,T}$, and $\partial h/\partial q|_{P,T} = \mu + T\,\partial s/\partial q|_{P,T}$.

2.3 Barodiffusion coefficient

Write $\mu(P, T, q)$ for the chemical potential as a function of pressure, temperature and composition. The barodiffusion coefficient is

$$\Gamma = -\frac{\partial\mu/\partial P}{\partial\mu/\partial q}, \tag{9}$$

with the derivatives at fixed (T, q) and fixed (P, T) respectively. It is independent of the choice of entropic variable, as $\mu(P, T, q)$ is. The heat flux below uses $\partial\vartheta/\partial q$ at fixed (P, T); for ϑ equal to the potential temperature of the fluid in Appendix A this derivative vanishes.

2.4 Equation of state

The equation of state is a general (nonlinear) binary fluid $h(P,\vartheta,q)$; Appendix A gives the explicit one currently used, with the constants left symbolic. Neither model assumes an ideal gas. The models require from the fluid, as functions of (P,ϑ,q): h, v, T, h_ϑ, h_q, c_p, $\partial_q\vartheta|_{P,T}$; as functions of (P,T,q): $\partial\mu/\partial P$ and $\partial\mu/\partial q$; and, for FC2D only, $P(v,\vartheta,q)$ and the sound speed c. AN2D only evaluates thermodynamic functions at the reference pressure $P_R(z)$.

3 Irreversible processes (common to both models)

3.1 Viscous stress

With constant dynamic viscosity η and the Stokes hypothesis (zero bulk viscosity),

$$\boldsymbol{\tau} = 2\eta\Big(\mathsf{S} - \tfrac{1}{3}(\boldsymbol{\nabla}\cdot\mathbf{u})\,\mathsf{I}\Big), \quad \mathsf{S} = \tfrac{1}{2}\big(\boldsymbol{\nabla}\mathbf{u} + \boldsymbol{\nabla}\mathbf{u}^\top\big), \qquad \mathbf{F} = \boldsymbol{\nabla}\cdot\boldsymbol{\tau} = \eta\Big(\tfrac{4}{3}\boldsymbol{\nabla}D + (\partial_z\omega,\,-\partial_x\omega)\Big), \tag{10}$$

where $D = \boldsymbol{\nabla}\cdot\mathbf{u}$ and $\mathbf{F}$ is the viscous force per unit volume [$\mathrm{N\,m^{-3}}$]. (With $\partial_y = 0$ and constant η, $\boldsymbol{\nabla}\cdot\boldsymbol{\tau} = \eta(\nabla^2\mathbf{u} + \frac{1}{3}\boldsymbol{\nabla}D)$ and $\nabla^2\mathbf{u} = \boldsymbol{\nabla}D + (\partial_z\omega, -\partial_x\omega)$.) The viscous dissipation per unit volume is

$$\rho\varepsilon = \boldsymbol{\tau} : \boldsymbol{\nabla}\mathbf{u} = \tfrac{2\eta}{3}\big(u_x^2 + w_z^2 + (u_x - w_z)^2\big) + \eta\,(u_z + w_x)^2 \;\geq 0, \tag{11}$$

and satisfies $\rho\varepsilon = \boldsymbol{\nabla}\cdot(\mathbf{u}\cdot\boldsymbol{\tau}) - \mathbf{u}\cdot\mathbf{F}$.

3.2 Heat and composition fluxes

Let $\mathbf{J}_q$ be the diffusive flux of solute (so $\rho\,\mathrm{D}q/\mathrm{D}t = -\boldsymbol{\nabla}\cdot\mathbf{J}_q$) and $\mathbf{J}_Q$ the total heat flux. The model uses the reduced heat flux $\mathbf{J}'_Q = \mathbf{J}_Q - \partial_q h|_{P,T}\,\mathbf{J}_q$ and the laws (no cross-diffusion)

$$\mathbf{J}_q = -\rho_f\,\kappa_q\,\big(\boldsymbol{\nabla}q - \Gamma\,\boldsymbol{\nabla}P_f\big), \tag{12}$$

$$\mathbf{J}'_Q = -\rho_f\,c_p\,\kappa_T\,\boldsymbol{\nabla}T. \tag{13}$$

For FC2D, $\rho_f = \rho$ and $P_f = P$. For AN2D, $\rho_f = \rho_R$, $P_f = P_R$, and all thermodynamic quantities (including Γ) are evaluated at $P = P_R(z)$; the pressure gradient is then $\boldsymbol{\nabla}P_R = -\rho_R g\,\hat{\mathbf{z}} = -\rho_R\boldsymbol{\nabla}\Phi$. Note $\boldsymbol{\nabla}\mu|_T = \partial_P\mu\,\boldsymbol{\nabla}P + \partial_q\mu\,\boldsymbol{\nabla}q = \partial_q\mu\,(\boldsymbol{\nabla}q - \Gamma\boldsymbol{\nabla}P)$, so $\mathbf{J}_q = -(\rho_f\kappa_q/\partial_q\mu)\,\boldsymbol{\nabla}\mu|_T$. An alternative "down-gradient" option, $\mathbf{J}_\vartheta = -\rho_f\kappa_T\boldsymbol{\nabla}\vartheta$, $\mathbf{J}_q = -\rho_f\kappa_q\boldsymbol{\nabla}q$, $\sigma_\vartheta = 0$, exists for comparison; it is not energy-conserving.

3.3 Evolution equation of ϑ and the production σ_ϑ

The enthalpy equation of a viscous binary fluid with heat flux $\mathbf{J}_Q$ is

$$\rho\frac{\mathrm{D}h}{\mathrm{D}t} = \frac{\mathrm{D}P}{\mathrm{D}t} - \boldsymbol{\nabla}\cdot\mathbf{J}_Q + \rho\varepsilon. \tag{14}$$

Substituting $\mathrm{D}h = v\,\mathrm{D}P + h_\vartheta\,\mathrm{D}\vartheta + h_q\,\mathrm{D}q$ from (3) and $\rho\,\mathrm{D}q/\mathrm{D}t = -\boldsymbol{\nabla}\cdot\mathbf{J}_q$:

$$\rho h_\vartheta\frac{\mathrm{D}\vartheta}{\mathrm{D}t} = -\boldsymbol{\nabla}\cdot\mathbf{J}_Q + h_q\,\boldsymbol{\nabla}\cdot\mathbf{J}_q + \rho\varepsilon = -\boldsymbol{\nabla}\cdot\big(\mathbf{J}_Q - h_q\mathbf{J}_q\big) - \mathbf{J}_q\cdot\boldsymbol{\nabla}h_q + \rho\varepsilon. \tag{15}$$

Define the flux of ϑ and use $h_\vartheta\,\boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta = \boldsymbol{\nabla}\cdot(h_\vartheta\mathbf{J}_\vartheta) - \mathbf{J}_\vartheta\cdot\boldsymbol{\nabla}h_\vartheta$:

$$\mathbf{J}_\vartheta = \frac{\mathbf{J}_Q - h_q\mathbf{J}_q}{h_\vartheta} = \frac{\mathbf{J}'_Q}{h_\vartheta} + \left.\frac{\partial\vartheta}{\partial q}\right|_{P,T}\mathbf{J}_q, \qquad \rho\frac{\mathrm{D}\vartheta}{\mathrm{D}t} = -\boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta + \sigma_\vartheta, \qquad h_\vartheta\,\sigma_\vartheta = \rho\varepsilon - \mathbf{J}_\vartheta\cdot\boldsymbol{\nabla}h_\vartheta - \mathbf{J}_q\cdot\boldsymbol{\nabla}h_q. \tag{16}$$

(The second form of $\mathbf{J}_\vartheta$ follows from (8).) In conservative form, $\partial_t(\rho\vartheta) + \boldsymbol{\nabla}\cdot(\rho\vartheta\mathbf{u}) = -\boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta + \sigma_\vartheta$ given mass conservation. This is exactly the system the models solve; it is also exactly energy-conserving (§7). Note $h_\vartheta\mathbf{J}_\vartheta + h_q\mathbf{J}_q = \mathbf{J}_Q$: the energy flux carried by the diffusive fluxes is the total heat flux.

3.4 Entropy production

The second law fixes the sign of the *entropy* production, not of σ_ϑ. From $T\,\mathrm{d}s = \mathrm{d}h - v\,\mathrm{d}P - \mu\,\mathrm{d}q$ and the enthalpy equation,

$$\rho T\frac{\mathrm{D}s}{\mathrm{D}t} = -\boldsymbol{\nabla}\cdot\mathbf{J}_Q + \mu\,\boldsymbol{\nabla}\cdot\mathbf{J}_q + \rho\varepsilon \quad\Longrightarrow\quad \rho\frac{\mathrm{D}s}{\mathrm{D}t} = -\boldsymbol{\nabla}\cdot\mathbf{J}_s + \sigma_s, \quad \mathbf{J}_s = \frac{\mathbf{J}_Q - \mu\mathbf{J}_q}{T}, \tag{17}$$

$$T\sigma_s = \rho\varepsilon - \mathbf{J}_s\cdot\boldsymbol{\nabla}T - \mathbf{J}_q\cdot\boldsymbol{\nabla}\mu. \tag{18}$$

(Expand $-\boldsymbol{\nabla}\cdot\mathbf{J}_Q/T + \mu\boldsymbol{\nabla}\cdot\mathbf{J}_q/T = -\boldsymbol{\nabla}\cdot\mathbf{J}_s - \mathbf{J}_s\cdot\boldsymbol{\nabla}T/T - \mathbf{J}_q\cdot\boldsymbol{\nabla}\mu/T$.) With $\mathbf{J}_Q = \mathbf{J}'_Q + (\mu + T\partial_q s|_{P,T})\mathbf{J}_q$ one has $\mathbf{J}_s = \mathbf{J}'_Q/T + \partial_q s|_{P,T}\mathbf{J}_q$, and by the Maxwell relation $\partial_q s|_{P,T}\,\boldsymbol{\nabla}T + \boldsymbol{\nabla}\mu = \boldsymbol{\nabla}\mu|_T$. Therefore

$$T\sigma_s = \rho\varepsilon - \frac{\mathbf{J}'_Q\cdot\boldsymbol{\nabla}T}{T} - \mathbf{J}_q\cdot\boldsymbol{\nabla}\mu|_T = \rho\varepsilon + \rho\,c_p\kappa_T\frac{|\boldsymbol{\nabla}T|^2}{T} + \rho\kappa_q\,\partial_q\mu\left|\boldsymbol{\nabla}q - \Gamma\boldsymbol{\nabla}P_f\right|^2 \geq 0, \tag{19}$$

by (11), (12), (13) and $\partial_q\mu > 0$ (for the fluid of the appendix $\partial_q\mu = R_sT/q$). Every term is non-negative.

Relation between σ_ϑ and σ_s. Since $s = s(\vartheta, q)$, $\rho\,\mathrm{D}s/\mathrm{D}t = s_\vartheta\,\rho\,\mathrm{D}\vartheta/\mathrm{D}t + s_q\,\rho\,\mathrm{D}q/\mathrm{D}t$ with s_ϑ, s_q from (7). Using $\mathbf{J}_s = s_\vartheta\mathbf{J}_\vartheta + s_q\mathbf{J}_q$ (which follows from $T\mathbf{J}_s = \mathbf{J}_Q - \mu\mathbf{J}_q = h_\vartheta\mathbf{J}_\vartheta + (h_q - \mu)\mathbf{J}_q$) and comparing the two expressions of $\rho\,\mathrm{D}s/\mathrm{D}t$:

$$T\sigma_s = h_\vartheta\,\sigma_\vartheta + T\big(\mathbf{J}_\vartheta\cdot\boldsymbol{\nabla}s_\vartheta + \mathbf{J}_q\cdot\boldsymbol{\nabla}s_q\big). \tag{20}$$

The correction vanishes if $\vartheta = s$ (then $s_\vartheta = 1$, $s_q = 0$). For a nonlinear $\vartheta(s, q)$ it does not (for potential temperature $s_\vartheta = c_{p0}/\vartheta$). Hence the *heating* $h_\vartheta\sigma_\vartheta$ that closes the energy budget is in general *not* sign-definite, even though the entropy production (19) is.

4 Fully compressible model (FC2D)

Prognostic variables: ρ, $\rho\vartheta$, ρq and $\mathbf{u}$. Pressure is diagnostic, $P = P(1/\rho, \vartheta, q)$.

$$\partial_t\rho = -\boldsymbol{\nabla}\cdot(\rho\mathbf{u}), \tag{21a}$$

$$\partial_t(\rho\vartheta) = -\boldsymbol{\nabla}\cdot(\rho\vartheta\,\mathbf{u}) - \boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta + \sigma_\vartheta, \tag{21b}$$

$$\partial_t(\rho q) = -\boldsymbol{\nabla}\cdot(\rho q\,\mathbf{u}) - \boldsymbol{\nabla}\cdot\mathbf{J}_q, \tag{21c}$$

$$\partial_t\mathbf{u} = -\vartheta\,\boldsymbol{\nabla}h_\vartheta - q\,\boldsymbol{\nabla}h_q - \boldsymbol{\nabla}\big(B + K + \Phi\big) - \boldsymbol{\omega}\times\mathbf{u} + \frac{\mathbf{F}}{\rho}, \tag{21d}$$

with $\boldsymbol{\omega}\times\mathbf{u} = (\omega w,\ -\omega u)$ and $B = h - \vartheta h_\vartheta - q h_q$ the Bernoulli potential (4). Here the gravitational force $-g\hat{\mathbf{z}} = -\boldsymbol{\nabla}\Phi$ has been combined with the gradients of B and K. By (5) the momentum equation is equivalent to $\partial_t\mathbf{u} + \boldsymbol{\omega}\times\mathbf{u} + \boldsymbol{\nabla}K = -v\boldsymbol{\nabla}P - g\hat{\mathbf{z}} + \mathbf{F}/\rho$. Sound waves are present.

5 Anelastic model (AN2D)

5.1 Reference state

The reference state is a density profile $\rho_R(z) > 0$ and the hydrostatic pressure

$$\frac{\mathrm{d}P_R}{\mathrm{d}z} = -\rho_R\,g, \qquad P_R(z_0) = P_{\mathrm{ref}}, \tag{22}$$

anchored at the height z_0 where the geopotential vanishes (`space.z0`; `pref` in the code). The anchor matters for the physics: the thermodynamic functions are evaluated at P_R, whereas z_0 alone only shifts the energy by a constant. Nothing else is specified: the thermodynamic state of the fluid is (ϑ, q), and $\rho_R(z)v(P_R, \vartheta, q) \neq 1$ in general; the mismatch is the buoyancy. The model replaces the full pressure by P_R in all thermodynamic functions: $h_\vartheta, h_q, B, T, c_p, \ldots$ below mean $h_\vartheta(P_R(z), \vartheta, q)$, etc.

5.2 Equations

Prognostic variables: $\rho_R\vartheta$, $\rho_R q$ and $\mathbf{u}$. There is no continuity equation; it is replaced by the constraint

$$\boldsymbol{\nabla}\cdot(\rho_R\mathbf{u}) = 0. \tag{23}$$

$$\partial_t(\rho_R\vartheta) = -\boldsymbol{\nabla}\cdot(\rho_R\vartheta\,\mathbf{u}) - \boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta + \sigma_\vartheta, \tag{24a}$$

$$\partial_t(\rho_R q) = -\boldsymbol{\nabla}\cdot(\rho_R q\,\mathbf{u}) - \boldsymbol{\nabla}\cdot\mathbf{J}_q, \tag{24b}$$

$$\partial_t\mathbf{u} = -\vartheta\,\boldsymbol{\nabla}h_\vartheta - q\,\boldsymbol{\nabla}h_q - \boldsymbol{\nabla}\big(B + K + \Phi\big) - \boldsymbol{\omega}\times\mathbf{u} + \frac{\mathbf{F}}{\rho_R} - \boldsymbol{\nabla}p, \tag{24c}$$

with $B = h - \vartheta h_\vartheta - q h_q$ evaluated at $P = P_R(z)$. The reduced pressure $p = (P - P_R)/\rho_R$ is the Lagrange multiplier enforcing (23): taking $\boldsymbol{\nabla}\cdot(\rho_R\,\cdot\,)$ of (24c) with $\partial_t\boldsymbol{\nabla}\cdot(\rho_R\mathbf{u}) = 0$ gives

$$\boldsymbol{\nabla}\cdot(\rho_R\boldsymbol{\nabla}p) = \boldsymbol{\nabla}\cdot(\rho_R\mathbf{R}), \qquad \mathbf{R} = -\vartheta\,\boldsymbol{\nabla}h_\vartheta - q\,\boldsymbol{\nabla}h_q - \boldsymbol{\nabla}\big(B + K + \Phi\big) - \boldsymbol{\omega}\times\mathbf{u} + \frac{\mathbf{F}}{\rho_R}, \tag{25}$$

with $\mathbf{n}\cdot\boldsymbol{\nabla}p$ determined by the wall condition below. Since P_R depends on z only, $\boldsymbol{\nabla}P_R = -\rho_R g\,\hat{\mathbf{z}}$ and the thermodynamic forcing in $\mathbf{R}$ is, using (5) at P_R,

$$-\vartheta\,\boldsymbol{\nabla}h_\vartheta - q\,\boldsymbol{\nabla}h_q - \boldsymbol{\nabla}\big(B + \Phi\big) = -v\,\boldsymbol{\nabla}P_R - g\hat{\mathbf{z}} = b\,\hat{\mathbf{z}}, \qquad b = g\left(\rho_R\,v(P_R,\vartheta,q) - 1\right), \tag{26}$$

so the anelastic model is driven by the buoyancy b alone: positive b means a parcel lighter than the reference. Three forcings are implemented:

- thermodynamic (default): the left-hand side above. In the continuum it equals $b\,\hat{\mathbf{z}}$ (its horizontal component is $v\,\partial_x P_R = 0$); discretely the horizontal component is only zero to truncation error;
- consistent buoyancy: $b = g(\rho_R v - 1)$ along $\hat{\mathbf{z}}$ only, identical to the above in the continuum;
- traditional buoyancy: $b = g(1 - 1/(\rho_R v)) = -g(\rho - \rho_R)/\rho_R$ with $\rho = 1/v$. This agrees with $g(\rho_R v - 1)$ only to first order in $\rho_R v - 1$.

6 Boundary conditions

Each side of the box is periodic or a wall. Periodic and wall sides may be mixed (e.g. periodic in x, walls in z). At a wall with inward normal $\mathbf{n}$ (unit, pointing into the fluid):

- *Impermeable:* $\mathbf{u}\cdot\mathbf{n} = 0$. In AN2D this is also the boundary condition of the pressure problem, $\rho_R(\mathbf{R} - \boldsymbol{\nabla}p)\cdot\mathbf{n} = 0$, i.e. no flux of $\rho_R\partial_t\mathbf{u}$ through the wall.
- *Tangential velocity* u_t (one of): free slip (no tangential stress), $\boldsymbol{\tau}_{nt} = 0$; no slip, $u_t = 0$; moving wall, $u_t = U_t$; prescribed stress $\eta\,\partial u_t/\partial n_{\text{out}} = \tau_w$ [$\mathrm{N\,m^{-2}}$], where $\partial/\partial n_{\text{out}}$ is the derivative along the outward normal $-\mathbf{n}$.
- *Composition* q and ϑ, one condition each: (i) a prescribed flux of ρq or $\rho\vartheta$ into the fluid $\mathbf{J}\cdot\mathbf{n}$ (the wall flux), (ii) a prescribed wall value of q or ϑ, or (iii) for ϑ only, a prescribed reduced heat flux into the fluid, $\mathbf{J}'_Q\cdot\mathbf{n} = F$ [$\mathrm{W\,m^{-2}}$]; then $\mathbf{J}_\vartheta\cdot\mathbf{n} = F/h_\vartheta + \partial_q\vartheta|_{P,T}\,\mathbf{J}_q\cdot\mathbf{n}$ by (16).

The wall pressure is needed to evaluate the boundary fluxes; it is extrapolated hydrostatically from the adjacent cell.

7 Properties to preserve under discretisation

7.1 Energy

Define the energy density per unit volume and the total energy

$$\mathcal{E} = \rho K + \mathcal{U}, \qquad \mathcal{U} = \begin{cases} \rho\,(h - Pv + \Phi) & \text{FC2D}, \\ \rho_R\,\big(h(P_R, \vartheta, q) + \Phi\big) - P_R & \text{AN2D}, \end{cases} \qquad E = \int \mathcal{E}\,\mathrm{d}A. \tag{27}$$

(For AN2D, ρK means $\rho_R K$.) The anelastic $\mathcal{U}$ is the compressible one with (ρ, P, v) replaced by $(\rho_R, P_R, 1/\rho_R)$, since $\rho_R(h - P_R v_R) = \rho_R h - P_R$; this is the sense in which the anelastic energy is traceable to the compressible energy.

Local energy equation, FC2D. Multiply (21d) by $\rho\mathbf{u}$ and use $\mathbf{u}\cdot(\boldsymbol{\omega}\times\mathbf{u}) = 0$:

$$\partial_t(\rho K) = K\,\partial_t\rho - \rho\vartheta\,\mathbf{u}\cdot\boldsymbol{\nabla}h_\vartheta - \rho q\,\mathbf{u}\cdot\boldsymbol{\nabla}h_q - \rho\mathbf{u}\cdot\boldsymbol{\nabla}(B + K + \Phi) + \mathbf{u}\cdot\mathbf{F}. \tag{28}$$

From (6), $\partial_t\mathcal{U} = (B+\Phi)\partial_t\rho + h_\vartheta\,\partial_t(\rho\vartheta) + h_q\,\partial_t(\rho q)$. Insert the tracer equations and add. The advective parts combine:

$$\begin{aligned} &K\partial_t\rho - \rho\mathbf{u}\cdot\boldsymbol{\nabla}K = -\boldsymbol{\nabla}\cdot(\rho\mathbf{u}K), \qquad -\rho\mathbf{u}\cdot\boldsymbol{\nabla}(B+\Phi) - (B+\Phi)\boldsymbol{\nabla}\cdot(\rho\mathbf{u}) = -\boldsymbol{\nabla}\cdot\big(\rho\mathbf{u}(B+\Phi)\big), \\ &-\rho\vartheta\mathbf{u}\cdot\boldsymbol{\nabla}h_\vartheta - h_\vartheta\boldsymbol{\nabla}\cdot(\rho\vartheta\mathbf{u}) = -\boldsymbol{\nabla}\cdot(h_\vartheta\rho\vartheta\mathbf{u}), \quad \text{likewise for } q, \end{aligned} \tag{29}$$

and $B + \vartheta h_\vartheta + q h_q = h$ gives the advective flux $\rho\mathbf{u}(K + h + \Phi)$. The irreversible parts are $\mathbf{u}\cdot\mathbf{F} - h_\vartheta\boldsymbol{\nabla}\cdot\mathbf{J}_\vartheta - h_q\boldsymbol{\nabla}\cdot\mathbf{J}_q + h_\vartheta\sigma_\vartheta = \mathbf{u}\cdot\mathbf{F} - \boldsymbol{\nabla}\cdot(h_\vartheta\mathbf{J}_\vartheta + h_q\mathbf{J}_q) + \rho\varepsilon$ by (16), and $\mathbf{u}\cdot\mathbf{F} + \rho\varepsilon = \boldsymbol{\nabla}\cdot(\mathbf{u}\cdot\boldsymbol{\tau})$. Hence

$$\partial_t\mathcal{E} = -\boldsymbol{\nabla}\cdot\Big[\rho\mathbf{u}\,(K + h + \Phi) + h_\vartheta\mathbf{J}_\vartheta + h_q\mathbf{J}_q - \mathbf{u}\cdot\boldsymbol{\tau}\Big]. \tag{30}$$

Local energy equation, AN2D. Now ρ_R is fixed and $\mathcal{U} = \mathcal{U}(\rho_R\vartheta, \rho_R q)$ with $\partial\mathcal{U}/\partial(\rho_R\vartheta) = h_\vartheta$, $\partial\mathcal{U}/\partial(\rho_R q) = h_q$ (differentiate $\rho_R h(P_R, \rho_R^{-1}m^\vartheta, \rho_R^{-1}m^q) - P_R$). The same steps give the advective terms $-\boldsymbol{\nabla}\cdot(h_\vartheta\rho_R\vartheta\mathbf{u} + h_q\rho_R q\mathbf{u})$, while $-\rho_R\mathbf{u}\cdot\boldsymbol{\nabla}(B + K + \Phi + p) = -\boldsymbol{\nabla}\cdot\big(\rho_R\mathbf{u}(B + K + \Phi + p)\big)$ uses the constraint (23). So (30) holds with $\rho \to \rho_R$ and the advective flux $\rho_R\mathbf{u}\,(K + h + \Phi + p)$. Note $\rho_R p = P - P_R$.

Wall budget. At impermeable walls $\mathbf{u}\cdot\mathbf{n} = 0$, so the advective flux in (30) does not contribute: for closed walls (no diffusive fluxes, no stress work) $\mathrm{d}E/\mathrm{d}t = 0$, and in general, with $\mathbf{n}$ the inward normal,

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \oint \Big(h_\vartheta\,\mathbf{J}_\vartheta + h_q\,\mathbf{J}_q - \mathbf{u}\cdot\boldsymbol{\tau}\Big)\cdot\mathbf{n}\,\mathrm{d}\ell = \oint \Big(\mathbf{J}_Q - \mathbf{u}\cdot\boldsymbol{\tau}\Big)\cdot\mathbf{n}\,\mathrm{d}\ell. \tag{31}$$

The integrand is the *wall input*: the total heat flux plus the work of the wall stress. For a heat-flux wall, $\mathbf{J}_Q\cdot\mathbf{n} = F + \partial_q h|_{P,T}\,\mathbf{J}_q\cdot\mathbf{n}$, so the wall delivers exactly F when no composition flows through it.

7.2 Other properties

- Mass (FC2D) and the tracers $\rho\vartheta$ (up to $\int\sigma_\vartheta$) and ρq change only through wall fluxes.
- The vortex force does no work: $\int\rho\,\mathbf{u}\cdot(\boldsymbol{\omega}\times\mathbf{u})\,\mathrm{d}A = 0$ (identically in the continuum).
- $\rho\varepsilon \geq 0$ and, for the consistent fluxes, the entropy production $T\sigma_s \geq 0$ (§3.4); the heating $h_\vartheta\sigma_\vartheta$ is given by (16).
- AN2D: $\boldsymbol{\nabla}\cdot(\rho_R\mathbf{u}) = 0$ at all times, and the pressure does no work.

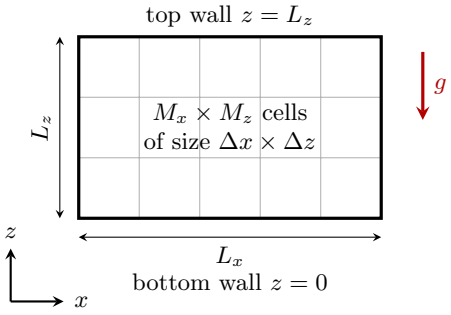
Part II
Numerical scheme

In Part II, ∂_x, ∂_z always denote the *discrete* derivatives defined in §8.3; continuous derivatives do not appear.

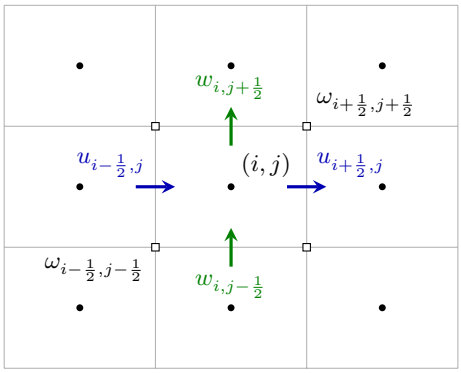
8 Grid, indices and operators

8.1 Geometry and C-grid

The box is divided into $M_x \times M_z$ uniform cells of size $\Delta x = L_x/M_x$, $\Delta z = L_z/M_z$ (Cartesian: no metric terms). Figure 1 shows the domain and the staggering of the variables.



(a) domain; x periodic or bounded, z bounded



(b) • cells (i,j): $m^c, \vartheta, q, h_\vartheta, h_q, B, K, P$;
$\rightarrow$ x-faces (I,j): u; $\uparrow$ z-faces (i,J): w; $\square$ vertices (I,J): ω

Figure 1: (a) The box and its cell grid, with gravity along $-z$. (b) The C-grid around cell (i,j): scalars at cell centres, u on x-faces, w on z-faces, vorticity at vertices. Cell indices i, j are integers; face and vertex (dual) indices I, J are half-integers.

8.2 Primal and dual indices

Primal indices i, j are integers and number the cells; **dual** indices I, J are half-integers and number the faces. The two x-faces of cell i are the *left* face $I = i - \frac{1}{2}$ and the *right* face $I = i + \frac{1}{2}$; the two z-faces of cell j are the *bottom* face $J = j - \frac{1}{2}$ and the *top* face $J = j + \frac{1}{2}$. Conversely, face I lies between cells $i = I - \frac{1}{2}$ (on its left) and $i = I + \frac{1}{2}$ (on its right), and face J between cells $j = J - \frac{1}{2}$ (below) and $j = J + \frac{1}{2}$ (above). A vertex is (I, J), the corner shared by the cells $(I \pm \frac{1}{2}, J \pm \frac{1}{2})$. For any index k (primal or dual) along x or z the coordinate is

$$x_k = (k - \tfrac{1}{2})\Delta x, \qquad z_k = (k - \tfrac{1}{2})\Delta z, \tag{32}$$

so that cell i is centred at $x_i = (i - \frac{1}{2})\Delta x$ and the face $I = i - \frac{1}{2}$ at $x_I = (i-1)\Delta x$. Interior cells are $i = 1, \dots, M_x$, $j = 1, \dots, M_z$. A bounded direction has $M + 1$ faces, $I = \frac{1}{2}, \frac{3}{2}, \dots, M_x + \frac{1}{2}$, the first and last being the walls (at $x = 0$ and $x = L_x$); a periodic direction has M faces, $I = \frac{1}{2}, \dots, M_x - \frac{1}{2}$, with $M_x + \frac{1}{2} \equiv \frac{1}{2}$. The z direction is always bounded. Indices outside the interior are *ghost* (halo) indices; the halo width $H \geq 1$ only needs to contain the stencils. In the code, the array index is $i + H$ for a cell index and $I + \frac{1}{2} + H$ for a face index (the code's face index n is the face on the left of cell n).

The location of each field follows from whether each index is primal or dual:

Location	Indices	Fields
cell centre	(i,j)	m^c, $\vartheta, q, T, h_\vartheta, h_q, c_p, \Gamma, B, P$, K, D, $\rho\varepsilon$, σ_ϑ, p, τ_{xx}, τ_{zz}
x-face	(I,j)	u, F_x, $\mathcal{G}_x$, $J_{\vartheta;x}$, $J_{q;x}$, $(\mathbf{u}\cdot\boldsymbol{\tau})_x$, S_x, ρ^u
z-face	(i,J)	w, F_z, $\mathcal{G}_z$, $J_{\vartheta;z}$, $J_{q;z}$, $(\mathbf{u}\cdot\boldsymbol{\tau})_z$, S_z, ρ^w
vertex	(I,J)	ω, τ_{xz}, ζ

The tracer array is $m^c_{i,j}$: $(m^\vartheta, m^q) = (\rho_R\vartheta, \rho_R q)$ in AN2D, and $(m^\rho, m^\vartheta, m^q) = (\rho, \rho\vartheta, \rho q)$ in FC2D (code `m[i,j,c]`).

8.3 Difference and average operators

For an array a with an index k along x (integer or half-integer), define

$$(\delta_x a)_k = a_{k+\frac{1}{2}} - a_{k-\frac{1}{2}}, \qquad (\mathcal{I}_x a)_k = \tfrac{1}{2}\big(a_{k+\frac{1}{2}} + a_{k-\frac{1}{2}}\big), \tag{33}$$

and identically $\delta_z, \mathcal{I}_z$ along z. The result lives at index k whose type (primal/dual) is the opposite of that of a: δ_x and $\mathcal{I}_x$ map x-faces to cells and cells to x-faces, e.g. $(\delta_x u)_{i,j} = u_{i+\frac{1}{2},j} - u_{i-\frac{1}{2},j}$ and $(\delta_x q)_{I,j} = q_{I+\frac{1}{2},j} - q_{I-\frac{1}{2},j}$. Averages across both directions compose, $\mathcal{I}_{xz} = \mathcal{I}_x\mathcal{I}_z$ (cells to vertices, or x-faces to z-faces). The *discrete derivatives*

$$\partial_x = \Delta x^{-1}\delta_x, \qquad \partial_z = \Delta z^{-1}\delta_z \tag{34}$$

are used from now on, as are $\mathcal{I}_x, \mathcal{I}_z$; formulas are written with these operators instead of explicit differences. A vertical profile f_j (such as ρ_R) is a field independent of i: $(\mathcal{I}_z f)_J = \frac{1}{2}(f_{J-\frac{1}{2}} + f_{J+\frac{1}{2}})$. All operators are the second-order centred scheme (`Centred2`). The identities they satisfy are stated in §14.

8.4 Halos and the state boundary conditions

Before every tendency evaluation the halo of the state is rebuilt from its interior (`state_bc!`); the interior is never changed except that wall-face normal velocities are set to 0. Directions are treated in the order x then z, each over the full extent (ghosts included) of the other direction, so corner ghosts are filled in the second pass.

Periodic direction. Indices are taken modulo M: $a_k \equiv a_{k+M}$.

Bounded direction, primal-type index (cell fields; the tangential velocity component): for $h = 1, \dots, H$, $a_{1-h} = \mathrm{g}_{\mathrm{lo}}(a_h, d)$ and $a_{M+h} = \mathrm{g}_{\mathrm{hi}}(a_{M+1-h}, d)$, with $d = (2h-1)\Delta$ the distance between the ghost and its mirror image.

Bounded direction, dual-type index (the wall-normal velocity): $a_{\frac{1}{2}} = a_{M+\frac{1}{2}} = 0$ and, for $h = 1, \dots, H$, $a_{\frac{1}{2}-h} = -a_{\frac{1}{2}+h}$ and $a_{M+\frac{1}{2}+h} = -a_{M+\frac{1}{2}-h}$ (odd about the wall).

Ghost rules g**.** For tracers m^c (and all derived thermodynamic fields) the rule is *zero gradient*, $\mathrm{g}(v, d) = v$. For the tangential velocity at a wall:

free slip	$v_{\mathrm{g}} = v$
no slip	$v_{\mathrm{g}} = -v$
moving wall, speed U_t	$v_{\mathrm{g}} = 2U_t - v$
prescribed stress τ_w	$v_{\mathrm{g}} = v + d\,\tau_w/\eta$

where $\tau_w = \eta\,\partial v/\partial n_{\mathrm{out}}$ with n_{out} the outward normal coordinate, so that the centred difference across the wall realises the condition. U_t is positive along $+x$ on bottom/top walls and along $+z$ on left/right walls.

Sampling of wall data. Wall data given as functions of the position along the wall are evaluated once, at the position of the field they apply to: tracer data (flux, heat flux, value) at the cell-centre coordinate x_i or z_j along the wall, and the tangential velocity or stress at the coordinate of that velocity component along the wall (z_J for w on a left/right wall, x_I for u on a bottom/top wall). Numbers are used as they are.

8.5 Evaluation ranges

In the kernels below, (i) cell fields are evaluated at interior cells; (ii) x-face fields at all interior x-faces I (including wall faces) and $j = 1, \dots, M_z$; (iii) z-face fields at $i = 1, \dots, M_x$ and all interior z-faces J; (iv) vertex fields at all interior I, J (including wall vertices). Derived fields (all but the state) needed at ghost cells receive the zero-gradient rule (or periodic wrap); they are read only to produce zero (differences across a wall) or in wall-face quantities that are overwritten or multiplied by the zero wall-normal velocity. Ghost values of the *state* do matter: they determine the vertices on the walls (ω, τ_{xz}, ζ).

9 The tendency computation

`tendencies!(dstate, scratch, model, state, t)` returns the time derivative of (m, u, w) as follows. Let ρ^c, ρ^u, ρ^w be the cell and face densities that weight the mass fluxes and the kinetic energy, and $v^u = 1/\rho^u$, $v^w = 1/\rho^w$:

	$\rho^c_{i,j}$	$\rho^u_{I,j}$	$\rho^w_{i,J}$
AN2D	$\rho_{R,j}$	$\rho_{R,j}$	$(\mathcal{I}_z\rho_R)_J$
FC2D	$m^\rho_{i,j}$	$(\mathcal{I}_x m^\rho)_{I,j}$	$(\mathcal{I}_z m^\rho)_{i,J}$

In every case $\rho^u = \mathcal{I}_x\rho^c$ and $\rho^w = \mathcal{I}_z\rho^c$. (AN2D uses the reference density in the halo cells $j = 0$, $j = M_z + 1$, §12; FC2D uses the zero-gradient ghost density.)

9.1 Step 0: halo of the state

As in §8.4.

9.2 Step 1: thermodynamics

At interior cells, then zero-gradient/periodic halo of every result. For each (i, j):

- AN2D: $\vartheta = m^\vartheta/\rho_{R,j}$, $q = m^q/\rho_{R,j}$, $P = P_{R,j}$.
- FC2D: $\rho = m^\rho$, $\vartheta = m^\vartheta/\rho$, $q = m^q/\rho$, $P = P(1/\rho, \vartheta, q)$ from the fluid.

Then $T, h_\vartheta, h_q, c_p, \partial_q\vartheta|_{P,T}$ are the fluid functions of (P, ϑ, q), $B = h(P, \vartheta, q) - \vartheta h_\vartheta - q h_q$ and, with $(\partial_P\mu, \partial_q\mu)$ the fluid functions of (P, T, q),

$$\Gamma = -\frac{\partial_P\mu}{\partial_q\mu} \qquad \text{(both models)}. \tag{35}$$

With the buoyancy options of AN2D, B is replaced by the buoyancy b: $b = g(\rho_R v - 1)$ (consistent) or $g(1 - 1/(\rho_R v))$ (traditional), with $v = v(P_{R,j}, \vartheta, q)$.

9.3 Step 2: kinematics

$$D = \partial_x u + \partial_z w \ \text{(cells)}, \qquad \omega = \partial_z u - \partial_x w \ \text{(vertices)}, \qquad K = \tfrac{1}{2}\big(\mathcal{I}_x(u^2) + \mathcal{I}_z(w^2)\big) \ \text{(cells)}, \tag{36}$$

where K is the average of the squares, not the square of the averages.

9.4 Step 3: advective fluxes

Centred, flux form, for each tracer c of m (and the mass flux $\mathcal{M}$):

$$\mathcal{G}^c_x = u\,\mathcal{I}_x m^c, \qquad \mathcal{G}^c_z = w\,\mathcal{I}_z m^c, \qquad \mathcal{M}_x = \rho^u u, \quad \mathcal{M}_z = \rho^w w. \tag{37}$$

In FC2D $\mathcal{M} = \mathcal{G}^\rho$ (since $\rho^u = \mathcal{I}_x m^\rho$); in AN2D $\mathcal{M}_x = \rho_R u$ is not stored.

9.5 Step 4: vortex force

The force $\boldsymbol{\omega}\times\mathbf{u} = (S_x, S_z) = (\omega w, -\omega u)$ is discretised in the energy-conserving form of Sadourny (1975):

$$\zeta = \frac{\omega}{\mathcal{I}_{xz}\rho^c}, \qquad U^v = \mathcal{I}_z\mathcal{M}_x, \quad W^v = \mathcal{I}_x\mathcal{M}_z \quad \text{(at vertices)}, \qquad S_x = \mathcal{I}_z(\zeta\, W^v), \quad S_z = -\mathcal{I}_x(\zeta\, U^v). \tag{38}$$

(For AN2D, $\mathcal{I}_{xz}\rho^c = \mathcal{I}_z\rho_R$.) Products are formed at vertices and then averaged to the faces. Sadourny's other scheme and Arakawa–Lamb's are also implemented in `CFBoxes` but are not specified here.

9.6 Step 5: viscous stress

$$\tau_{xx} = 2\eta\big(\partial_x u - \tfrac{1}{3}D\big), \quad \tau_{zz} = 2\eta\big(\partial_z w - \tfrac{1}{3}D\big) \ \text{(cells)}, \qquad \tau_{xz} = \eta\big(\partial_z u + \partial_x w\big) \ \text{(vertices)}. \tag{39}$$

The energy flux $\mathbf{u}\cdot\boldsymbol{\tau}$ and the force $\mathbf{F}$ (in the rotational form of (10)) at the faces are

$$(\mathbf{u}\cdot\boldsymbol{\tau})_x = u\,\mathcal{I}_x\tau_{xx} + \mathcal{I}_z\big((\mathcal{I}_x w)\,\tau_{xz}\big), \qquad (\mathbf{u}\cdot\boldsymbol{\tau})_z = w\,\mathcal{I}_z\tau_{zz} + \mathcal{I}_x\big((\mathcal{I}_z u)\,\tau_{xz}\big), \tag{40}$$

$$F_x = \eta\big(\partial_z\omega + \tfrac{4}{3}\partial_x D\big), \qquad F_z = \eta\big(-\partial_x\omega + \tfrac{4}{3}\partial_z D\big). \tag{41}$$

At wall vertices $\mathcal{I}_x w$ and $\mathcal{I}_z u$ contain the ghost values, so the wall stress work is that of the tangential wall velocity. The dissipation at interior cells is

$$\rho\varepsilon = \partial_x(\mathbf{u}\cdot\boldsymbol{\tau})_x + \partial_z(\mathbf{u}\cdot\boldsymbol{\tau})_z - \mathcal{I}_x(uF_x) - \mathcal{I}_z(wF_z). \tag{42}$$

9.7 Step 6: diffusive fluxes

The density in the fluxes is the face density of the table above, ρ^u at x-faces and ρ^w at z-faces. Coefficients at faces are $\mathcal{I}_x$ or $\mathcal{I}_z$ of the cell values. The pressure gradient in the barodiffusion term is that of P_f, the pressure of the model: $P_f = P$ in FC2D (a cell field) and $P_f = P_R$ in AN2D, a profile with

$$\partial_x P_R = 0, \qquad \partial_z P_R = -g\,\mathcal{I}_z\rho_R = -g\rho^w \quad \text{exactly (§12)}. \tag{43}$$

Then, at x-faces,

$$J_{q;x} = -\rho^u\kappa_q\big(\partial_x q - (\mathcal{I}_x\Gamma)\,\partial_x P_f\big), \tag{44}$$

$$J_{\vartheta;x} = -\frac{\rho^u\,\kappa_T\,(\mathcal{I}_x c_p)\,\partial_x T}{\mathcal{I}_x h_\vartheta} + (\mathcal{I}_x\partial_q\vartheta)\,J_{q;x}, \tag{45}$$

and the same at z-faces with ρ^w, $\mathcal{I}_z$, ∂_z (with $\partial_z P_f$). The diffusive heating at faces is

$$H_x = -\big(J_{\vartheta;x}\,\partial_x h_\vartheta + J_{q;x}\,\partial_x h_q\big), \qquad H_z = -\big(J_{\vartheta;z}\,\partial_z h_\vartheta + J_{q;z}\,\partial_z h_q\big). \tag{46}$$

At wall faces the fluxes are overwritten by the boundary conditions (§10); at a wall face the ghost values of h_ϑ, h_q equal the cell values, so $H = 0$ there. The production of $\rho\vartheta$ at interior cells (consistent fluxes) is

$$\sigma_\vartheta = \frac{\rho\varepsilon + \mathcal{I}_x H_x + \mathcal{I}_z H_z}{h_\vartheta}, \tag{47}$$

the discrete form of (16). With the down-gradient option: $J_{q;x} = -\rho^u\kappa_q\partial_x q$, $J_{\vartheta;x} = -\rho^u\kappa_T\partial_x\vartheta$ (and in z), $\sigma_\vartheta = 0$.

9.8 Step 7: momentum tendencies

Before the pressure correction, at interior x-faces (I, j) and z-faces (i, J) (the wall faces are set in step 10):

$$\dot{u} = -v^u \Big[(\mathcal{I}_x m^\vartheta)\, \partial_x h_\vartheta + (\mathcal{I}_x m^q)\, \partial_x h_q \Big] - \partial_x \big(B + K \big) - S_x + v^u F_x, \tag{48}$$

$$\dot{w} = -v^w \Big[(\mathcal{I}_z m^\vartheta)\, \partial_z h_\vartheta + (\mathcal{I}_z m^q)\, \partial_z h_q \Big] - \partial_z \big(B + K + \Phi \big) - S_z + v^w F_z, \tag{49}$$

with $\Phi_j = g(z_j - z_0)$, for which $\partial_z \Phi = g$ exactly. Here $B = h - \vartheta h_\vartheta - q h_q$ is the Bernoulli potential of step 1; m^ϑ, m^q are the tracer densities (m^1, m^2 in AN2D, m^2, m^3 in FC2D), so that $v^u \mathcal{I}_x m^\vartheta$ is the discrete ϑ at the x-face. In AN2D, with a buoyancy option ($B \to b$ computed in step 1), instead

$$\dot{u} = -\partial_x K - S_x + v^u F_x, \qquad \dot{w} = \mathcal{I}_z b - \partial_z K - S_z + v^w F_z. \tag{50}$$

The arrays are wrapped in periodic directions.

9.9 Step 8 (AN2D only): pressure correction

The reduced pressure p, and $\dot{u} \leftarrow \dot{u} - \partial_x p$, $\dot{w} \leftarrow \dot{w} - \partial_z p$ at all faces; see §11.

9.10 Step 9: tracer tendencies

At interior cells,

$$\dot{m}^c = -\partial_x \big(\mathcal{G}^c_x + J^c_x \big) - \partial_z \big(\mathcal{G}^c_z + J^c_z \big) + \sigma^c, \tag{51}$$

with $J^\rho = 0$, $\sigma^\rho = \sigma^q = 0$, $J^\vartheta = J_\vartheta$, $J^q = J_q$, $\sigma^\vartheta = \sigma_\vartheta$ (the mass m^ρ exists only in FC2D).

9.11 Step 10: tendency boundary conditions

Set to zero: $\dot{m}$ in the halo, $\dot{u}, \dot{w}$ in the halo, $\dot{u}$ on x-walls and $\dot{w}$ on z-walls. The state is not otherwise modified.

10 Wall conditions on the fluxes

After step 6, on each wall face the fluxes J_ϑ, J_q are replaced, at every position along the wall, by the following. Let $n = +1$ (lower wall: $I = \frac{1}{2}$ or $J = \frac{1}{2}$) or $n = -1$ (upper: $I = M_x + \frac{1}{2}$ or $J = M_z + \frac{1}{2}$) be the inward normal, i_c the adjacent cell ($i = 1$ or M_x, $j = 1$ or M_z), $d = \Delta/2$ the half-cell distance; fluxes are in the coordinate direction. The wall state $(\rho_w, P_w, \partial P_{f;w})$ (wall density, wall pressure and the wall value of the gradient ∂P_f used in the composition flux) is:

	AN2D	FC2D
x-wall	$\rho_{R,j},\ P_{R,j},\ 0$	$\rho_{i_c},\ P_{i_c},\ 0$
z-wall	$(\mathcal{I}_z \rho_R)_J,\ P_{R,j} + n \rho_{R,j} g\, d,\ -(\mathcal{I}_z \rho_R)_J g$	$\rho_{i_c},\ P_{i_c} + n \rho_{i_c} g\, d,\ -\rho_{i_c} g$

- **Composition.** Prescribed flux F of ρq: $J_q = nF$. Prescribed wall value q_w: $J_q = -\rho_w \kappa_q \big(n(q_{i_c} - q_w)/d - \Gamma_{i_c}\, \partial P_{f;w} \big)$.
- ϑ. Prescribed flux F of $\rho\vartheta$: $J_\vartheta = nF$. Prescribed heat flux F (reduced heat flux into the fluid): $J_\vartheta = nF/h_{\vartheta i_c} + \partial_q \vartheta_{i_c} J_q$. Prescribed value ϑ_w: $J_\vartheta = -\rho_w\, c_{p;i_c} \kappa_T\, n (T_{i_c} - T_w)/(d\, h_{\vartheta i_c}) + \partial_q \vartheta_{i_c} J_q$ with $T_w = T(P_w, \vartheta_w, q_{i_c})$.

Because the ghost values of h_ϑ and h_q equal the cell values, the energy delivered by a wall is $h_{\vartheta i_c} J_\vartheta + h_{q_{i_c}} J_q$ (with the sign convention of (31)), which for a heat-flux wall is $F + \partial_q h|_{P,T} J_q n$.

11 Pressure projection and the Poisson solver (AN2D)

11.1 The discrete problem

After step 7 let $(\dot{u}, \dot{w})$ be the tendencies. First set $\dot{u} = 0$ on x-walls and $\dot{w} = 0$ on z-walls. Define the weighted divergence and the operator L on interior cells:

$$r = \rho_R\, \partial_x \dot{u} + \partial_z\big(\rho^w \dot{w}\big), \tag{52}$$

$$Lp = \rho_R\, \partial_x \partial_x p + \partial_z\big(\rho^w\, \partial_z p\big), \tag{53}$$

and find p at interior cells with $Lp = r$. In (53) the ghost values of p are the zero-gradient ones, $p_{0,j} = p_{1,j}$, $p_{M_x+1,j} = p_{M_x,j}$ for bounded x (periodic wrap otherwise) and $p_{i,0} = p_{i,1}$, $p_{i,M_z+1} = p_{i,M_z}$ in z, which makes the gradient of p vanish on the walls. Written out, $(Lp)_{i,j} = \rho_{R,j}(p_{i+1,j} - 2p_{i,j} + p_{i-1,j})/\Delta x^2 + \big[\rho^w_{j+\frac{1}{2}}(p_{i,j+1} - p_{i,j}) - \rho^w_{j-\frac{1}{2}}(p_{i,j} - p_{i,j-1})\big]/\Delta z^2$.

Projection. Let $Gp = (\partial_x p, \partial_z p)$ at faces (zero on wall faces). Then, directly from the definitions, $L = \mathrm{div}_\rho \circ G$ where $\mathrm{div}_\rho(a, b) = \rho_R \partial_x a + \partial_z(\rho^w b)$. Hence $\mathrm{div}_\rho\big((\dot{u}, \dot{w}) - Gp\big) = r - Lp = 0$, and the corrected tendency has zero wall-normal component. In the code the gradients at wall faces are 0 because the ghost value of p equals the cell value, and the correction is applied at all faces.

Properties of L. (i) L is symmetric for the unweighted inner product $\langle a, b\rangle = \sum_{i,j} a_{i,j} b_{i,j}$: the x-part is $\rho_{R,j}$ times the same symmetric second difference in each row, the z-part is a symmetric conductance matrix with weights ρ^w. (ii) L is negative semi-definite: by summation by parts (§14), $\langle p, Lp\rangle = -\sum_{I,j} \rho_{R,j}(\partial_x p)^2 - \sum_{i,J} \rho^w_J (\partial_z p)^2 \le 0$, sums over interior faces. (iii) The kernel is the constants: $\langle p, Lp\rangle = 0$ forces all differences to vanish. (iv) $\sum_{i,j}(Lp)_{i,j} = 0$ for all p, hence $Lp = r$ is solvable iff $\sum_{i,j} r_{i,j} = 0$. This holds: $\sum_i r_{i,j}$ telescopes in x (periodic wrap or zero wall velocity), and $\sum_j$ telescopes in z since $\dot{w} = 0$ at both walls. The solution is unique up to a constant, fixed by the gauge below.

11.2 Diagonalisation in x

Write $L = \rho_R \otimes A_x + \mathrm{Id} \otimes T_z$ where A_x is the x-second difference (acting within a row) and T_z the tridiagonal z-operator (acting within a column). Since T_z does not depend on i and $\rho_{R,j}$ does not depend on i, any orthogonal basis diagonalising A_x with eigenvalues λ_m decouples L into independent tridiagonal systems in j, one per mode m.

- *Periodic x*: eigenvectors $e^{2\pi \mathrm{i}\, m(i-1)/M_x}$, $m = 0, \dots, M_x - 1$; $(e^{\mathrm{i}\theta(i+1)} - 2e^{\mathrm{i}\theta i} + e^{\mathrm{i}\theta(i-1)})/\Delta x^2 = -\frac{4\sin^2(\theta/2)}{\Delta x^2} e^{\mathrm{i}\theta i}$ with $\theta = 2\pi m/M_x$, so $\lambda_m = -\big(2\sin(\pi m/M_x)/\Delta x\big)^2$, which is also the value for $M_x - m$ (conjugate modes).
- *Bounded x* (Neumann ghosts $p_0 = p_1$, $p_{M+1} = p_M$): eigenvectors $\cos\big(\pi m(i - \frac{1}{2})/M_x\big)$, $m = 0, \dots, M_x - 1$ (discrete cosine transform of type II). Check: both ghost conditions hold for these (e.g. $p_0 = \cos(\pi m/2M_x) = p_1$), and the three-term sum equals $2\cos(\cdot)\big(\cos(\pi m/M_x) - 1\big)$, so $\lambda_m = -\big(2\sin(\pi m/(2M_x))/\Delta x\big)^2$.

In both cases $\lambda_0 = 0$ and $\lambda_m < 0$ otherwise.

Transforms as implemented (FFTW conventions).

- Periodic x: forward = real-to-complex FFT (`rfft`) giving $\lfloor M_x/2 \rfloor + 1$ complex coefficients $\hat{r}_m = \sum_{i=1}^{M_x} r_i\, e^{-2\pi \mathrm{i}\, m(i-1)/M_x}$; backward = complex-to-real (`brfft`, unnormalised), which reconstructs the real signal from the half spectrum assuming Hermitian symmetry, so that

backward$\circ$forward $= M_x \cdot \mathrm{id}$. The real and imaginary parts of the half spectrum are stored interleaved, giving $2(\lfloor M_x/2 \rfloor + 1)$ *real rows*, each a real tridiagonal system with the same λ_m. The imaginary parts of $m = 0$ (and of $m = M_x/2$ for even M_x) do not enter the backward transform.

- Bounded x: forward = DCT-II (`REDFT10`): $\hat{r}_m = 2\sum_{i=1}^{M_x} r_i \cos(\pi m(i - \frac{1}{2})/M_x)$; backward = DCT-III (`REDFT01`), with backward$\circ$forward $= 2M_x \cdot \mathrm{id}$.

Let s be M_x (periodic) or $2M_x$ (bounded). The code multiplies the right-hand side by $1/s$ when forming r, so that after the backward transform the result is p itself.

11.3 Tridiagonal solve in z (Thomas algorithm)

For each real row (mode m) the system at levels $j = 1, \ldots, M_z$ is

$$a_j \hat{p}_{j-1} + b_j \hat{p}_j + c_j \hat{p}_{j+1} = \hat{r}_j, \qquad a_j = \frac{\rho^w_{j-\frac{1}{2}}}{\Delta z^2}, \quad c_j = \frac{\rho^w_{j+\frac{1}{2}}}{\Delta z^2}, \quad b_j = \lambda_m \rho_{R,j} - a_j - c_j, \tag{54}$$

with $a_1 = 0$ and $c_{M_z} = 0$ (no flux through the walls). The factorisation is precomputed once per model (it depends only on ρ_R and Δ): pivots $\mathrm{pv}_1 = b_1$, $\mathrm{pv}_j = b_j - a_j\, f_{j-1}$ with $f_j = c_j/\mathrm{pv}_j$, and $\mathrm{ip}_j = 1/\mathrm{pv}_j$. The solve is:

$$y_1 = \hat{r}_1\, \mathrm{ip}_1, \qquad y_j = (\hat{r}_j - a_j y_{j-1})\, \mathrm{ip}_j \quad (j = 2, \ldots, M_z), \tag{55}$$

$$\hat{p}_{M_z} = y_{M_z}, \qquad \hat{p}_j = y_j - f_j \hat{p}_{j+1} \quad (j = M_z - 1, \ldots, 1). \tag{56}$$

11.4 Gauge

For $\lambda_m = 0$ ($m = 0$; both its real rows) the matrix is singular: $b_j = -a_j - c_j$, and the rows sum to zero because $c_j = a_{j+1}$ and $a_1 = c_{M_z} = 0$. The factorisation is computed level by level as above, but for these rows the level-one values are overwritten right after level one is processed, $f_1 \leftarrow 0$ and $\mathrm{ip}_1 \leftarrow 0$, so that $\mathrm{pv}_2 = b_2$. The solve then gives $y_1 = 0$, $y_2 = \hat{r}_2/b_2$, …, and back substitution gives $\hat{p}_1 = y_1 - f_1 \hat{p}_2 = 0$. In effect the level-one equation is discarded and $\hat{p}_1 = 0$ is imposed, so *the horizontal mean of p on the bottom level is zero.* Because the rows sum to zero, the discarded equation is the negative of the sum of the retained ones, and is therefore satisfied exactly when $\sum_j \hat{r}_j = 0$ for the mean mode, which is property (iv). The total pressure is $P = P_R + \rho_R p$, defined up to this gauge. The user may add a constant to p after the solve (`gauge_offset`, default 0, so that the horizontal mean of p on the bottom level is the offset). It changes the pressure but no tendency, and no term of the energy budget except the fluxes $\rho_R \Sigma \mathbf{u}$ and $(P - P_R)\mathbf{u}$, which change by the constant times the mass flux; their divergences do not, since $\nabla\cdot(\rho_R \mathbf{u}) = 0$ discretely (tested).

11.5 Assembly

The steps in `pressure!`: (1) zero the wall-normal $\dot{u}, \dot{w}$ on the walls; (2) form r/s at interior cells; (3) forward transform each level j in x; (4) Thomas solve each row; (5) backward transform; (6) copy into the interior of p and set the halo of p by zero gradient/periodic wrap; (7) $\dot{u} \leftarrow \dot{u} - \partial_x p$ at all x-faces, $\dot{w} \leftarrow \dot{w} - \partial_z p$ at all z-faces. In step 7 the wall faces receive 0 from the zero-gradient halo, so the corrected tendency has zero wall-normal component.

12 Anelastic reference state and initialisation (AN2D)

12.1 Reference state

The reference state is held at the cell centres of the *full* vertical axis, halo included, $j = 1 - H, \ldots, M_z + H$:

- $\rho_{R,j}$: a number ρ_0; or a function $\rho_R(z)$ evaluated at the cell centres z_j (also in the halo, i.e. the function is extrapolated); or a vector of M_z values, extended into the halo by repeating its end values.
- $P_{R,j}$: hydrostatic, $P_{R,j} - P_{R,j-1} = -g\Delta z\, \mathcal{I}_z \rho_R$ between the centres $j-1$ and j (the density of the face between them is the average of the two cells), up to a constant fixed by the anchor. The pressure at z_0 is that of the piecewise-linear interpolant of the cell-centre values $P_{R,j}$, taken in the segment between two interior cells that contains z_0 (linearly extrapolated from the nearest segment if z_0 is outside the box), and it equals P_{ref}.

Consequently $\partial_z P_R = -g\,\mathcal{I}_z \rho_R = -g\rho^w$ exactly: the discrete hydrostatic balance uses the face density ρ^w that weights the vertical mass flux and the Poisson operator. FC2D has no reference state; its balanced rest density (`balanced_density`) has the pressure P_{ref} at z_0 in the same sense, the pressure at the top cell being found by secant iterations on the interpolated pressure at z_0. The specific volume is not stored; face values of ρ_R are $\mathcal{I}_z \rho_R$.

12.2 Initial state

Given $\vartheta(x,z)$ and $q(x,z)$: $m^{\vartheta}_{i,j} = \rho_{R,j}\vartheta(x_i,z_j)$, $m^{q}_{i,j} = \rho_{R,j} q(x_i,z_j)$ at every cell including the halo (then replaced by the halo rule, §8.4). The velocity is either zero or given by face arrays. A discretely divergence-free velocity is obtained from a mass streamfunction $\psi(x,z)$ [$\mathrm{kg\,m^{-1}s^{-1}}$] evaluated at the vertices (x_I, z_J):

$$\rho^u u = \partial_z \psi, \qquad \rho^w w = -\partial_x \psi, \tag{57}$$

so that $\mathrm{div}(\rho^u u, \rho^w w) = \partial_x\partial_z\psi - \partial_z\partial_x\psi = 0$ identically. The wall-normal velocity vanishes if ψ is constant along the wall. Energy conservation of AN2D requires a state on this constraint manifold (§14).

13 Time integration

Both models are advanced with the classical fourth-order Runge–Kutta scheme in the following (low-storage) form, with f the tendency map and Δt the step; all state arrays are updated at every point including the halo, which is then rebuilt at each evaluation of f:

$$k_0 = f(\mathbf{s}^n), \qquad \mathbf{s}_1 = \mathbf{s}^n + \tfrac{\Delta t}{2} k_0, \qquad k_1 = f(\mathbf{s}_1), \tag{58}$$

$$\mathbf{s}_2 = \mathbf{s}_1 - \tfrac{\Delta t}{2} k_0 + \tfrac{\Delta t}{2} k_1, \qquad k_2 = f(\mathbf{s}_2), \tag{59}$$

$$\mathbf{s}_3 = \mathbf{s}_2 - \tfrac{\Delta t}{2} k_1 + \Delta t\, k_2, \qquad k_3 = f(\mathbf{s}_3), \tag{60}$$

$$\mathbf{s}^{n+1} = \mathbf{s}_3 + \tfrac{\Delta t}{6} k_0 + \tfrac{\Delta t}{3} k_1 - \tfrac{2\Delta t}{3} k_2 + \tfrac{\Delta t}{6} k_3, \tag{61}$$

i.e. $\mathbf{s}_2 = \mathbf{s}^n + \frac{\Delta t}{2} k_1$, $\mathbf{s}_3 = \mathbf{s}^n + \Delta t\, k_2$ and $\mathbf{s}^{n+1} = \mathbf{s}^n + \frac{\Delta t}{6}(k_0 + 2k_1 + 2k_2 + k_3)$. A five-stage Kinnmark–Gray scheme is also available and is not specified here. Time is advanced in slices of fixed length τ_s (the output interval). At the start of each slice

$$\Delta t = \frac{\tau_s}{\lceil \tau_s / \min(C\sqrt{8}/\omega_{\max},\, \tau_s,\, \Delta t_{\max}) \rceil}, \tag{62}$$

where $C \le 1$ is the Courant number, $\sqrt{8}$ the stability limit of RK4 on the imaginary axis, $\Delta t_{\max}$ a user cap, and $\omega_{\max}$ is evaluated from the state at the start of the slice:

$$\omega_{\max} = \max\frac{|u|}{\Delta x} + \max\frac{|w|}{\Delta z} + 4\max(\nu, \kappa_T, \kappa_q)\Big(\frac{1}{\Delta x^2} + \frac{1}{\Delta z^2}\Big) + N_{\max}\,\Big[+ 2c_{\max}\sqrt{\tfrac{1}{\Delta x^2} + \tfrac{1}{\Delta z^2}}\ \mathrm{FC2D}\Big], \tag{63}$$

with maxima over the interior, $\nu = \frac{4}{3}\eta/\min\rho$ (ρ_R for AN2D, m^ρ for FC2D), and $N_{\max}$ the largest buoyancy frequency over the interior z-faces $J = \frac{3}{2}, \dots, M_z - \frac{1}{2}$:

$$N^2 = \frac{g}{v}\Big[\frac{\partial v}{\partial \vartheta}\,\partial_z\vartheta + \frac{\partial v}{\partial q}\,\partial_z q\Big], \tag{64}$$

with $v, \partial v/\partial\vartheta, \partial v/\partial q$ the fluid functions at the pressure $\mathcal{I}_z P$ (P_R for AN2D) and the face averages $\mathcal{I}_z\vartheta, \mathcal{I}_z q$; $N_{\max} = \sqrt{\max N^2}$ (and 0 if $N^2 < 0$ everywhere). $c_{\max}$ is the largest sound speed $c(v, \vartheta, q)$ over cells.

14 Discrete energy conservation

14.1 Summation by parts

All sums run over interior indices. Let $\Sigma_c = \sum_{i,j}$ over cells, Σ_u over interior x-faces (including walls), Σ_w over interior z-faces, Σ_v over vertices. For a at x-faces and c at cells, rearranging the sums gives, in a bounded direction,

$$\sum_{i=1}^{M_x} c_i\,(\delta_x a)_i = -\sum_{I=\frac{3}{2}}^{M_x-\frac{1}{2}} a_I\,(\delta_x c)_I + c_{M_x} a_{M_x+\frac{1}{2}} - c_1 a_{\frac{1}{2}}, \tag{65}$$

and without the wall terms in a periodic direction. Likewise

$$\sum_{i=1}^{M_x} c_i\,(\mathcal{I}_x a)_i = \sum_{I=\frac{3}{2}}^{M_x-\frac{1}{2}} a_I\,(\mathcal{I}_x c)_I + \tfrac{1}{2}\big(c_1 a_{\frac{1}{2}} + c_{M_x} a_{M_x+\frac{1}{2}}\big). \tag{66}$$

If a vanishes on the walls, as does u, a mass flux, or any flux set to zero there, both reduce to

$$\Sigma_c\, c\,\delta_x a = -\Sigma_u\, a\,\delta_x c, \qquad \Sigma_c\, c\,\mathcal{I}_x a = \Sigma_u\, a\,\mathcal{I}_x c, \tag{67}$$

and the same holds in z (for w-type fields) and between vertices and faces. Applying the second identity to $\rho^c\mathcal{I}_x(u^2)$ with $\rho^u = \mathcal{I}_x\rho^c$, and likewise in z:

$$\Sigma_c\,\rho^c K = \tfrac{1}{2}\Big(\Sigma_u\,\rho^u u^2 + \Sigma_w\,\rho^w w^2\Big). \tag{68}$$

The kinetic energy is thus weighted by the same face densities ρ^u, ρ^w as the mass fluxes, the vortex force and the pressure correction.

14.2 The discrete energy

$$E_d = \Delta x\Delta z\,\Sigma_c\Big[\rho^c K + \mathcal{U}\Big], \qquad \mathcal{U} = \begin{cases} \rho\,(h - Pv + \Phi_j) & \text{FC2D}, \\ \rho_{R,j}(h(P_{R,j}, \vartheta, q) + \Phi_j) - P_{R,j} & \text{AN2D}, \end{cases} \quad \Phi_j = g(z_j - z_0). \tag{69}$$

An additive constant in Φ changes E_d by a constant (since $\Sigma_c\rho^c$ is conserved: trivially for AN2D, and by step 9 for FC2D). $\mathcal{U}$ is a pointwise function of the cell values $m^c_{i,j}$ with the derivatives

$$\frac{\partial\mathcal{U}}{\partial m^\vartheta}\bigg| = h_\vartheta, \quad \frac{\partial\mathcal{U}}{\partial m^q}\bigg| = h_q, \quad \frac{\partial\mathcal{U}}{\partial m^\rho}\bigg| = B + \Phi_j \text{ (FC2D)}, \tag{70}$$

by (6) (FC2D) and its AN2D analogue, provided the fluid supplies h, v, h_ϑ, h_q satisfying (3) exactly (thermodynamic consistency of the equation of state, which is assumed).

14.3 Result

Take the consistent fluxes and heating of steps 5–6 and the energy-conserving vortex force of step 4. For FC2D, or for AN2D with a state satisfying $\operatorname{div}(\rho^u u, \rho^w w) = 0$ in every cell, the semi-discrete energy obeys

$$\frac{\mathrm{d}E_d}{\mathrm{d}t} = \sum_{\text{wall faces } f} \ell_f \, n_f \Big[h_{\vartheta i_c} J_{\vartheta;f} + h_{q_{i_c}} J_{q;f} - (\mathbf{u}\cdot\boldsymbol{\tau})_f \Big], \tag{71}$$

where ℓ_f is the length of the wall face (Δz for an x-wall, Δx for a z-wall), $n_f = \pm 1$ the inward normal, i_c the adjacent cell, J_f the wall-face flux in the coordinate direction (as set in §10) and $(\mathbf{u}\cdot\boldsymbol{\tau})_f$ the wall-face value of step 5. This is the discrete counterpart of (31). In particular $\dot{E}_d = 0$, to round-off, for periodic or closed boundaries (no wall flux, no tangential wall motion or stress).

14.4 Derivation

Let $V = \Delta x \Delta z$ and $\mathcal{M} = (\rho^u u, \rho^w w)$ the face mass fluxes, $\operatorname{div}\mathcal{M} = \partial_x \mathcal{M}_x + \partial_z \mathcal{M}_z$. The chain rule on (68) and on $\mathcal{U}$ gives

$$\frac{\dot{E}_d}{V} = \Sigma_u \, \mathcal{M}_x \dot{u} + \Sigma_w \, \mathcal{M}_z \dot{w} + \Sigma_c \, \dot{m}^\rho \, K \;\; \text{(FC2D only)} + \Sigma_c \big[h_\vartheta \dot{m}^\vartheta + h_q \dot{m}^q + (B+\Phi)\dot{m}^\rho \big] \tag{72}$$

(the $\dot{m}^\rho$ terms are absent for AN2D). The $\dot{m}^\rho K$ term arises because the weights ρ^u, ρ^w depend on time in FC2D: $\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\Sigma_c \rho^c \mathcal{I}_x(u^2) = \Sigma_c \dot{\rho}^c \, \frac{1}{2}\mathcal{I}_x(u^2) + \Sigma_u \rho^u u \dot{u}$ by (67). Insert $\dot{u}, \dot{w}$ from (48), (49) (and the pressure correction for AN2D), using $\rho^u v^u = 1$, and $\dot{m}^c$ from (51).

(a) Thermodynamic force against the tracer fluxes. The thermodynamic part of $\mathcal{M}_x \dot{u}$ is $-\sum_{c=\vartheta,q} u\,(\mathcal{I}_x m^c)\,\partial_x \Pi^c = -\sum_c \mathcal{G}^c_x \, \partial_x \Pi^c$ with $\Pi^\vartheta = h_\vartheta$, $\Pi^q = h_q$, by the definition of $\mathcal{G}^c_x$ (step 3). By (67) ($\mathcal{G}^c_x = 0$ on walls since $u = 0$ there), $-\Sigma_u \, \mathcal{G}^c_x \, \partial_x \Pi^c = +\Sigma_c \, \Pi^c \, \partial_x \mathcal{G}^c_x$; the same in z. The advective part of $\Pi^c \dot{m}^c$ is $-\Pi^c(\partial_x \mathcal{G}^c_x + \partial_z \mathcal{G}^c_z)$. They cancel exactly.

(b) Mass terms, FC2D. Multiplying the terms $-\partial_x(B+K)$ and $-\partial_z(B+K+\Phi)$ of (48)–(49) by the mass fluxes and summing by parts gives $+\Sigma_c (B+K+\Phi)\operatorname{div}\mathcal{M}$; here $\partial_z \Phi = g$ exactly, and the x-derivative of Φ is zero since Φ is independent of i. The mass tendency is $\dot{m}^\rho = -\operatorname{div}\mathcal{M}$ (step 9 with $J^\rho = \sigma^\rho = 0$, $\mathcal{M} = \mathcal{G}^\rho$), so the corresponding energy terms $\Sigma_c (B+\Phi+K)\dot{m}^\rho = -\Sigma_c (B+\Phi+K)\operatorname{div}\mathcal{M}$ cancel them exactly.

(c) Mass terms, AN2D. $\dot{m}^\rho$ is absent, but by hypothesis $\operatorname{div}\mathcal{M} = 0$, so $\Sigma_c (B+K+\Phi)\operatorname{div}\mathcal{M} = 0$ and the same terms vanish. The pressure correction contributes $-\Sigma_u \mathcal{M}_x \partial_x p - \Sigma_w \mathcal{M}_z \partial_z p = \Sigma_c \, p \operatorname{div}\mathcal{M} = 0$.

(d) Vortex force. It enters as $-\Sigma_u \mathcal{M}_x S_x - \Sigma_w \mathcal{M}_z S_z$. By the second identity of (67) (vertex-to-face averages are adjoint to face-to-vertex averages, $\mathcal{M} = 0$ on the walls) with step 4,

$$\begin{aligned} \Sigma_u \mathcal{M}_x S_x &= \Sigma_u \mathcal{M}_x \mathcal{I}_z(\zeta W^v) = \Sigma_v \, \zeta \, W^v \, \mathcal{I}_z \mathcal{M}_x = \Sigma_v \, \zeta \, W^v U^v, \\ \Sigma_w \mathcal{M}_z S_z &= -\Sigma_w \mathcal{M}_z \mathcal{I}_x(\zeta U^v) = -\Sigma_v \, \zeta \, U^v W^v, \end{aligned} \tag{73}$$

which cancel. The vertex sums run over interior vertices because wall vertices contribute nothing: on a z-wall $W^v = \mathcal{I}_x \mathcal{M}_z = 0$, and on an x-wall $U^v = \mathcal{I}_z \mathcal{M}_x = 0$ (the wall-normal mass flux vanishes), so every term evaluated at a wall vertex is multiplied by zero. The identity holds for any ζ: the discrete vortex force does no work regardless of ω or ρ.

(e) Viscous work and heating. The viscous terms contribute $\Sigma_u \rho^u u\, v^u F_x + \Sigma_w \rho^w w\, v^w F_z = \Sigma_u\, uF_x + \Sigma_w\, wF_z$. Sum (42) over cells: the first two terms telescope by (65) with $c = 1$, and the last two are $\Sigma_c \mathcal{I}_x(uF_x) = \Sigma_u\, uF_x$ and $\Sigma_c \mathcal{I}_z(wF_z) = \Sigma_w\, wF_z$ by (66) ($u = w = 0$ on walls). Hence

$$\Sigma_u\, uF_x + \Sigma_w\, wF_z = -\Sigma_c\, \rho\varepsilon + (\text{wall terms of } \partial_x(\mathbf{u}\cdot\boldsymbol{\tau})_x + \partial_z(\mathbf{u}\cdot\boldsymbol{\tau})_z). \tag{74}$$

The wall terms are $\sum_j \big[(\mathbf{u}\cdot\boldsymbol{\tau})_{x;M_x+\frac{1}{2},j} - (\mathbf{u}\cdot\boldsymbol{\tau})_{x;\frac{1}{2},j}\big]/\Delta x$ and the z analogue, i.e. $-n_f(\mathbf{u}\cdot\boldsymbol{\tau})_f/\Delta_\perp$ per wall face, with $\Delta_\perp$ the spacing normal to the wall. Multiplied by V and using $V/\Delta_\perp = \ell_f$ they give the stress-work term of (71). (Periodic directions have no wall term.)

(f) Diffusion and the production. The tracer tendency contains $-\partial_x J^c_x - \partial_z J^c_z$ for $c = \vartheta, q$ and $+\sigma_\vartheta$. By (65) with boundary terms,

$$-\Sigma_c\, \Pi^c\big(\partial_x J^c_x + \partial_z J^c_z\big) = +\Sigma_{\text{interior faces}} J^c\, \partial\Pi^c + \Sigma_{\text{wall faces}} \frac{n_f\, \Pi^c_{i_c} J^c_f}{\Delta_\perp}, \tag{75}$$

where $J^c\partial\Pi^c$ denotes the product of a face flux with the face gradient (in x or z as appropriate), the first sum is over interior faces, and the wall term is $-\Pi^c_M J^c_{M+\frac{1}{2}} + \Pi^c_1 J^c_{\frac{1}{2}}$ in each direction, using the cell value $\Pi^c_{i_c}$ of the adjacent cell. Multiplied by V this gives $\ell_f n_f \Pi^c_{i_c} J^c_f$. The production gives, by (47) and (66) (using $H = 0$ on wall faces): $\Sigma_c\, h_\vartheta \sigma_\vartheta = \Sigma_c\, \rho\varepsilon + \Sigma_{\text{interior faces}} H = \Sigma_c \rho\varepsilon - \Sigma_{\text{interior faces}}\big(J_\vartheta \partial h_\vartheta + J_q \partial h_q\big)$. Adding the diffusion and production, the interior-face terms cancel exactly, leaving $\Sigma_c \rho\varepsilon + \Sigma_{\text{walls}} n_f (h_{\vartheta i_c} J_\vartheta + h_{q i_c} J_q)/\Delta_\perp$.

Summing (a)–(f): all interior terms cancel, $-\Sigma_c\rho\varepsilon$ from (e) against $+\Sigma_c\rho\varepsilon$ from (f), and the remaining wall terms, multiplied by V, are exactly (71).

Remarks. (1) The derivation uses only: the staggering and weights of steps 2–9, the exact thermodynamic relations (70), the zero-gradient halo of h_ϑ, h_q, and $u = w = 0$ on walls; no property of ζ, of the Poisson solution beyond the constraint, or of the time step. (2) For AN2D the constraint $\mathrm{div}_\rho \mathbf{u} = 0$ is a hypothesis on the *state*, whereas the Poisson solve enforces $\mathrm{div}_\rho \dot{\mathbf{u}} = 0$ on the *tendency*. The former is preserved by the latter, both semi-discretely and under Runge–Kutta stepping: every stage tendency satisfies $\mathrm{div}_\rho k = 0$ whatever the stage state is, so $\mathrm{div}_\rho \mathbf{s}^{n+1} = \mathrm{div}_\rho \mathbf{s}^n$ up to round-off. A state built from the streamfunction (or any state with $\mathrm{div}_\rho \mathbf{u} = 0$) therefore stays on the manifold. Energy, however, is conserved only semi-discretely; Runge–Kutta stepping is not conservative. (3) With the down-gradient fluxes ($\sigma_\vartheta = 0$), (f) fails: the interior face terms and the viscous heating $\rho\varepsilon$ are not returned as heat, so energy is lost. (4) With a buoyancy option the force is not $-v\boldsymbol{\nabla}P$ in discrete form and (a)–(c) do not hold exactly.

15 Discrete dissipation and entropy production

What is exact. (i) $\rho\varepsilon \geq 0$ in every cell: with F the discrete divergence of $\boldsymbol{\tau}$ (step 5), the discrete product rule $\partial_x(u\,\mathcal{I}_x\tau) - \mathcal{I}_x(u\,\partial_x\tau) = \tau\,\partial_x u$ (immediate from the definitions of $\delta_x, \mathcal{I}_x$) applied to the normal stresses, and the same identity at the vertices applied to τ_{xz}, give

$$\begin{aligned} \rho\varepsilon &= \tau_{xx}\partial_x u + \tau_{zz}\partial_z w + \mathcal{I}_{xz}\big(\tau_{xz}(\partial_z u + \partial_x w)\big) \\ &= \tfrac{2\eta}{3}\Big[(\partial_x u)^2 + (\partial_z w)^2 + (\partial_x u - \partial_z w)^2\Big] + \eta\,\mathcal{I}_{xz}\big((\partial_z u + \partial_x w)^2\big) \;\geq 0. \end{aligned} \tag{76}$$

The force F computed in step 5 is exactly the discrete divergence of the discrete stress: $\partial_x\tau_{xx} + \partial_z\tau_{xz} = \eta\big[\tfrac{4}{3}\partial_x\partial_x u + \partial_z\partial_z u + \tfrac{1}{3}\partial_x\partial_z w\big] = F_x$ because ∂_x, ∂_z commute on the uniform grid, and likewise for F_z. (ii) The heat returned to $\rho\vartheta$ in a cell, $h_\vartheta\sigma_\vartheta = \rho\varepsilon + \mathcal{I}_x H_x + \mathcal{I}_z H_z$, makes the *global* energy budget close exactly (§14); in particular the total dissipated kinetic energy $\Sigma_c\rho\varepsilon \geq 0$ reappears as internal energy.

What is not exact. The continuous entropy production (19) is a sum of non-negative terms. Its discrete analogue is not guaranteed non-negative: the face heating $H = -(J_\vartheta \partial h_\vartheta + J_q \partial h_q)$ combines face-averaged coefficients with cell differences, and by (20) the heating $h_\vartheta \sigma_\vartheta$ differs from $T\sigma_s$ by $-T(\mathbf{J}_\vartheta \cdot \boldsymbol{\nabla} s_\vartheta + \mathbf{J}_q \cdot \boldsymbol{\nabla} s_q)$, which has no sign even in the continuum when $\vartheta \neq s$. The scheme therefore guarantees exact energy closure and $\rho\varepsilon \geq 0$, and is a second-order consistent discretisation of (16) (hence of (19)), but it does not guarantee a discrete entropy inequality. Moreover, since the advective fluxes are centred, the quantity $\Sigma_c \rho s$ is not discretely conserved by advection either.

16 Differences between AN2D and FC2D

	AN2D	FC2D
Prognostic m	$(\rho_R\vartheta, \rho_R q)$	$(\rho, \rho\vartheta, \rho q)$
Density	fixed profile $\rho_R(z)$	continuity equation (step 9, $c = \rho$)
Pressure	p from Poisson solve (§11)	diagnostic $P(1/\rho, \vartheta, q)$
Thermodynamics at	$P_{R,j}$	local $P_{i,j}$
Constraint	$\mathrm{div}(\rho^u u, \rho^w w) = 0$ every stage	none; acoustic modes
Face densities ρ^u, ρ^w	$\rho_{R,j}$, $\mathcal{I}_z\rho_R$	$\mathcal{I}_x m^\rho$, $\mathcal{I}_z m^\rho$
Vortex-force density	profile ρ_R (vertex: $\mathcal{I}_z\rho_R$)	cell array m^ρ (vertex: $\mathcal{I}_{xz} m^\rho$)
Barodiffusion pressure gradient ∂P_f	∂P_R: 0 on x-faces, $-g\rho^w$ on z-faces	∂P of the diagnostic pressure
Forcing options	thermodynamic or buoyancy	thermodynamic only
Potential energy $\mathcal{U}$	$\rho_R(h(P_R) + \Phi) - P_R$	$\rho(h - Pv + \Phi)$
Time step	advection, diffusion, buoyancy	plus sound
Initial state	given ρ_R, ϑ, q; divergence-free $\mathbf{u}$ from ψ	balanced density in discrete hydrostatic balance (not specified here)

All other kernels (steps 0–7, 9, 10, the wall conditions on fluxes and the vortex force) are the same code, mirrored by hand between the two packages.

A Nonlinear binary equation of state

With ϑ the potential temperature (reference pressure P_0), reference values (P_0, T_0, q_0) and constants $v_0, \alpha_T, \alpha_q, \alpha_P, \gamma, \alpha_{TT}, c_{p0}$ and $R_s = R/M_s$ (values set per experiment):

$$v = v_0\Big[1 + \alpha_T\big(1 + \gamma(P - P_0)\big)(\vartheta - T_0) - \alpha_q(q - q_0) - \alpha_P(P - P_0) + \tfrac{1}{2}\alpha_{TT}(\vartheta - T_0)^2\Big], \tag{77}$$

$$h = c_{p0}(\vartheta - T_0) + v_0\Big[1 + \alpha_T\big(1 + \tfrac{1}{2}\gamma(P - P_0)\big)(\vartheta - T_0) - \alpha_q(q - q_0) - \tfrac{1}{2}\alpha_P(P - P_0) + \tfrac{1}{2}\alpha_{TT}(\vartheta - T_0)^2\Big](P - P_0), \tag{78}$$

$$h_\vartheta = c_{p0} + v_0\Big[\alpha_T\big(1 + \tfrac{1}{2}\gamma(P - P_0)\big) + \alpha_{TT}(\vartheta - T_0)\Big](P - P_0), \qquad T = \frac{\vartheta\, h_\vartheta}{c_{p0}}, \tag{79}$$

$$h_q = -v_0\alpha_q\,(P - P_0), \qquad s = c_{p0}\ln\frac{\vartheta}{T_0} + s_{\mathrm{mix}}(q), \quad s_{\mathrm{mix}} = -R_s\, q(\ln q - 1), \tag{80}$$

$$\mu = h_q - s'_{\mathrm{mix}}(q)\, T, \tag{81}$$

$$\left.\frac{\partial\mu}{\partial P}\right|_{T,q} = -v_0\alpha_q, \qquad \left.\frac{\partial\mu}{\partial q}\right|_{P,T} = -s''_{\mathrm{mix}}(q)\, T = \frac{R_s T}{q}, \tag{82}$$

$$c^2 = -v^2 \Big/ \left.\frac{\partial v}{\partial P}\right|_{\vartheta,q}. \tag{83}$$

Hence $\Gamma = -\partial_P \mu / \partial_q \mu = v_0 \alpha_q \, q/(R_s T)$, and with this fluid $\partial\vartheta/\partial q|_{P,T} = 0$. Also $s_\vartheta = h_\vartheta/T = c_{p0}/\vartheta$ and $s_q = s'_{\mathrm{mix}}(q)$, consistent with (7). The enthalpy is defined up to an additive constant (reference enthalpy), which does not affect the dynamics.